Calculus I — Limits & Derivatives

limits

Which technique? — try them in this order

1. **Substitute.** If you get a number, that is the limit. Always try this first.
2. Got $\frac{0}{0} \Rightarrow$ **algebraic techniques**: factor and cancel, multiply by the conjugate, or clear a complex fraction.
3. Still $\frac{0}{0}$ or $\frac{\infty}{\infty} \Rightarrow$ **L'Hôpital's rule**. Differentiate top and bottom separately.
4. Trapped between two bounds, or a $\sin\frac{1}{x}$ factor $\Rightarrow$ **squeeze theorem**.
5. $x \to \pm\infty$ on a rational function $\Rightarrow$ **compare leading exponents**.
6. A piecewise split, or $|x|$ at its zero $\Rightarrow$ **one-sided limits**. They must agree.

A number over zero is not $\frac{0}{0}$ — that is an infinite limit or a vertical asymptote, and no technique above applies.

discontinuities

Which kind? — test in this order

A **removable** — the two-sided limit exists but misses $f(c)$, or $f(c)$ is undefined. A hole. Fix it by redefining one point.

B **jump** — both one-sided limits exist but disagree. Typical of a piecewise split.

C **infinite** — a one-sided limit runs to $\pm\infty$. A vertical asymptote.

THE TEST
1. **Compute** both one-sided limits.
2. **Do they disagree?** That is a jump.
3. **Is either infinite?** That is an infinite discontinuity.
4. **Do they agree but miss $f(c)$?** That is removable.

related rates

THE PROCESS
1. **Draw** the picture and name every changing quantity.
2. **Write the equation** that ties them together. Geometry, not calculus.
3. **Differentiate both sides with respect to t.** Every variable gets a $\frac{d}{dt}$ by the chain rule.
4. **Substitute the instant** — the given values and rates. Substitute only now, never in step 2.
5. **Solve** for the unknown rate.

Which equation ties them together?

the picture shows	the equation
a right triangle, ladder shadow	$a^2+b^2=c^2$
similar triangles	a ratio of matching sides
a cone or a cylinder filling	V formula, then eliminate a variable with a ratio
an angle changing	$\tan\theta=\frac{\text{opp}}{\text{adj}}$
area or volume growing	the area or volume formula

optimization (one variable)

THE PROCESS
1. **Name** the quantity to maximize or minimize. Write it as an equation.
2. **Find the constraint** — the second relationship in the problem.
3. **Substitute the constraint** to reduce to one variable.
4. **State the domain.** A length cannot be negative.
5. **Differentiate, set to zero**, and solve.
6. **Test the candidates** and the endpoints, then answer the question asked.

Calculus II — Integration

integration by parts (IBP)

IBP vs u-sub — which one?

- u-sub: a composition *and* its derivative is present elsewhere.
- IBP: a product of unrelated functions, no inside-derivative.

area between curves

THE PROCESS
1. **Sketch** both curves.
2. **Find every intersection.** Solve $f(x)=g(x)$.
3. **Choose the slice direction** — vertical (dx) or horizontal (dy).
4. **Integrate the gap**: $(\text{top}-\text{bottom})\,dx$, or $(\text{right}-\text{left})\,dy$.

Step 3 decoded — three region shapes

A dx — one curve stays on top across the whole interval. $\int_a^b (\text{top}-\text{bottom})\,dx$

B dy — the region reads better left-to-right. Solve both curves for x first, then $\int_c^d (\text{right}-\text{left})\,dy$

C **curves cross inside** the interval $\Rightarrow$ split at each crossing. The top curve changes there.

THE THREE THINGS YOU SOLVE
1. **Intersections** — set $f(x)=g(x)$ and solve. These are the limits, or the split points.
2. **Which curve is on top** — test one interior point. Do not guess from the formulas.
3. **Solve for x** when you slice horizontally: $y=x^2 \Rightarrow x=\sqrt{y}$.

volumes of revolution

THE PROCESS
1. **Sketch** the region and draw the axis.
2. **Choose the slice** — perpendicular to the axis (disc or washer), or parallel to it (shell).
3. **Write each radius** as a distance from the axis.
4. **Integrate.**

Step 2 decoded — disc, washer or shell?

A **disc** — slice $\perp$ axis, and the region touches the axis: $V=\pi\int R^2$

B **washer** — slice $\perp$ axis, but a gap sits between region and axis: $V=\pi\int (R_{\text{out}}^2-R_{\text{in}}^2)$

C **shell** — slice $\parallel$ axis: $V=2\pi\int (\text{radius})(\text{height})$

Discs and washers integrate in the axis variable; shells integrate in the other one. If solving the curve for the other variable is ugly, switch method rather than fight it.

THE THREE THINGS YOU SOLVE
1. **Intersections** — where the region starts and stops.
2. **The radius.** About $y=k$ the radius is $|k-y|$, not y. Draw the arrow from axis to curve.
3. **Solve for the other variable** when the slice direction needs it: $y=\sqrt{x} \Rightarrow x=y^2$.

work (springs & pumping)

THE PROCESS — ALL THREE KINDS
1. **Slice** the object or the fluid.
2. **Write the force** on one slice.
3. **Write the distance** that same slice moves. Different slices move different distances.
4. **Integrate** force $\times$ distance.

Step 1 decoded — three kinds of work

A **spring** — force varies, distance is the same. Force $=kx$, so $W=\int_a^b kx\,dx$

B **lifting a cable or chain** — force varies as rope leaves the load. Slice the rope.

C **pumping fluid** — every slice has the same force law but a different distance to travel. Slice the fluid horizontally.

THE THREE THINGS YOU SOLVE
1. **The constant k** — from one given force and stretch: $F=kx$.
2. **The weight of one slice** — density $\times$ volume for fluid, mass per length for rope.
3. **The distance that slice moves** — measure from the slice to its destination, not to the origin.

hydrostatic force

THE PROCESS
1. Draw the plate and choose a depth variable.
2. Take a horizontal strip at depth h.
3. Write the strip width as a function of that depth.
4. Integrate $\rho g \times$ depth $\times$ width.

improper integrals

THE PROCESS
1. Find the bad point — an infinite bound, or where the integrand blows up.
2. Replace it with a limit variable.
3. Integrate normally.
4. Take the limit. If it is finite, the integral converges. If not, it diverges.

Step 1 decoded — three kinds

A **Type 1** — an infinite bound: $\int_1^{\infty}$
B **Type 2, endpoint** — the integrand blows up at a or b: $\int_0^1 \frac{dx}{\sqrt{x}}$
C **Type 2, interior** — the blow-up sits inside the interval. Split there first, and both halves must converge.

C is the trap: $\int_{-1}^{1} \frac{dx}{x^2}$ looks routine and is divergent. Check the integrand for a zero denominator before you integrate anything.

Vector Calculus

cross product (3D only)

What is it for? — four jobs

A a vector **perpendicular to two others** $\Rightarrow \vec{u}\times\vec{v}$, then divide by its magnitude for a unit normal
B **area** $\Rightarrow$ parallelogram $=|\vec{u}\times\vec{v}|$; triangle $=\frac{1}{2}|\vec{u}\times\vec{v}|$
C **volume** of a parallelepiped $\Rightarrow |\vec{u}\cdot(\vec{v}\times\vec{w})|$ (triple scalar product)
D **torque** $\Rightarrow \vec{r}\times\vec{F}$

All four run the same 3×3 determinant. Only what you do with the result changes. The cross product exists in 3D only — a 2D problem must be lifted with a z-component of 0.

lines & planes in space

Given this, want that — the lookup

you are given	the move
line through two points	direction $\vec{v}=P_2-P_1$
line $\perp$ to a plane	direction $\vec{v}=\vec{n}$
plane from a point + normal	$\vec{n}\cdot\langle x-x_0, y-y_0, z-z_0\rangle=0$
plane through three points	$\vec{n}=(P_2-P_1)\times(P_3-P_1)$
plane from a point + two vectors	$\vec{n}=\vec{v}_1\times\vec{v}_2$
plane $\perp$ to a line	$\vec{n}=\vec{v}$
distance, point to line	$\frac{\lvert\vec{PM}\times\vec{v}\rvert}{\lvert\vec{v}\rvert}$
distance, point to plane	$\frac{\lvert ax_0+by_0+cz_0-d\rvert}{\lvert\vec{n}\rvert}$
distance between skew lines	$\frac{\lvert(P_2-P_1)\cdot(\vec{v}_1\times\vec{v}_2)\rvert}{\lvert\vec{v}_1\times\vec{v}_2\rvert}$
angle between two planes	the angle between $\vec{n}_1$ and $\vec{n}_2$
line where two planes meet	direction $=\vec{n}_1\times\vec{n}_2$

THE WHOLE SECTION IN TWO OBJECTS
A **line** needs a point and a direction. A **plane** needs a point and a normal.
Every problem above is one question: which of those two am I missing, and what produces it?
The answer is almost always a **difference** of points, or a **cross product**.

Two nonparallel vectors span the plane's directions, so $\vec{v}_1\times\vec{v}_2$ is perpendicular to both and therefore normal to the plane. This is the three-point case with one step removed — given points, you build the two vectors first.

Lines — three equivalent forms

Through point $P_0(x_0,y_0,z_0)$ with direction $\vec{v}=\langle a,b,c\rangle$:

- **Vector**: $\vec{r}=\vec{r}_0+t\vec{v}$ ($\vec{r}_0=\langle x_0,y_0,z_0\rangle$; t slides the point along the line)
- **Parametric**: $x=x_0+ta$, $y=y_0+tb$, $z=z_0+tc$
- **Symmetric** (solve each for t): $\frac{x-x_0}{a}=\frac{y-y_0}{b}=\frac{z-z_0}{c}$

If a direction component is 0, that coordinate is frozen — write it separately, for example, $b=0$: $\frac{x-x_0}{a}=\frac{z-z_0}{c}$, $y=y_0$.

Through **two points** P,Q: direction $\vec{v}=\vec{PQ}$ (tip $-$ tail), then use either point as P_0. The **segment** PQ only: $\vec{r}(t)=(1-t)\vec{p}+t\vec{q}$, $0\le t\le 1$.

vector-valued functions (space curves)

THE PROCESS
1. Write two letters — $\vec{r}_1(t)$ and $\vec{r}_2(s)$. Reusing t silently asks the collide question.
2. Set the components equal — that is three equations.
3. Solve two of them for t and s.
4. Check the third. If it fails, the curves do not intersect.
5. Report the point, not the parameters — substitute back.

calculus of vector-valued functions

THE PROCESS — WORK COMPONENTWISE
1. Differentiate or integrate each component on its own. The vector structure never interferes.
2. Add the constant vector $\vec{C}$ after every integration.
3. Use products carefully — the dot and cross rules follow the ordinary product rule, but the cross product does not commute, so keep the order.

arc length

The routine — same four moves every time

Differentiate componentwise $\rightarrow$ square & sum the slots $\rightarrow$ simplify under the root $\rightarrow$ integrate.

- Never integrate the raw root. Problems are rigged so the radicand collapses to a constant (trig slots — Pythagorean identity) or a perfect square ($(A+B)^2$ — check: middle term $=2AB$). If it won't collapse, hunt the missed identity.
- **"Set up only"** $\Rightarrow$ stop at the integral.

motion in space

Which direction are you travelling? — two jobs

A **given position** $\vec{r}(t)\Rightarrow$ differentiate down: $\vec{v}=\vec{r}\,'$, $\vec{a}=\vec{r}\,''$, speed $=|\vec{v}|$
B **given acceleration** plus initial conditions $\Rightarrow$ integrate up: $\vec{v}=\int\vec{a}\,dt$, then $\vec{r}=\int\vec{v}\,dt$. Each integration adds a constant vector, fixed by $\vec{v}(0)$ and $\vec{r}(0)$.

The constant of integration is a vector, not a scalar. Three components, three conditions.

Speed is a scalar, velocity is a vector. Speed is $|\vec{v}|$, and a magnitude discards direction — so an object can have constant speed while its velocity changes every instant.

multivariable limits & continuity

Which job is this? — three outcomes

A **substitute and it works** $\Rightarrow$ that is the limit. Always try this first.
B **show the limit does NOT exist** $\Rightarrow$ find two paths giving different values. One counterexample finishes it.
C **prove the limit DOES exist** $\Rightarrow$ paths can never prove existence. Use the squeeze theorem or convert to polar.

B and C need opposite tools, so decide which you are doing before you start. Trying paths for hours cannot settle case C, and a squeeze argument is wasted effort in case B.

partial derivatives

READING ANY RATE

Units $= \dfrac{\text{units of the output}}{\text{units of the input that varies}}$ — read them straight off the problem statement.

Sign = increasing or decreasing as that one input grows.

f_x holds y fixed. It describes motion parallel to the x-axis and says nothing about any other direction.

tangent planes & linear approximation

How is the surface written? — two forms, one machine

- **A** **a graph,** $z=f(x,y) \Rightarrow z=f(a,b)+f_x(a,b)(x-a)+f_y(a,b)(y-b)$
- **B** **a level surface,** $F(x,y,z)=c \Rightarrow \nabla F$ is the normal: $\nabla F\cdot\langle x-x_0, y-y_0, z-z_0\rangle=0$

A is B in disguise. Put the graph in level-surface form as $F=f(x,y)-z$, and $\nabla F=\langle f_x, f_y, -1\rangle$ — which is the coefficient pattern in A. Use B whenever z cannot be isolated, as in $x^2+4y^2+z^2=18$. The level-surface case is worked in the gradient section below.

Evaluate the partials at the point before building the plane. Leaving f_x as a formula produces a surface, not a plane.

multivariable chain rule

Which tree? — three shapes

- **A** **one independent variable** (x,y both depend on t) $\Rightarrow$ a total derivative: $\dfrac{dz}{dt}=\dfrac{\partial z}{\partial x}\dfrac{dx}{dt}+\dfrac{\partial z}{\partial y}\dfrac{dy}{dt}$
- **B** **two independent variables** (x,y depend on u,v) $\Rightarrow$ one partial per independent variable, each a full sum
- **C** **implicit** — no explicit formula, only $F(x,y)=0 \Rightarrow \dfrac{dy}{dx}=-\dfrac{F_x}{F_y}$

THE PROCESS

1. **Draw the tree.** Put the output on top and the independent variables at the bottom.
2. **Count the paths** from the output down to the variable you differentiate by.
3. **Multiply along each path**, then **add the paths**.
4. **Substitute** the inner functions only at the end.

Build the formula before you compute anything. The tree tells you how many terms to expect, so a missing term becomes visible immediately.

directional derivatives & the gradient

What is being asked? — four questions

- **A** **rate in a given direction** $\Rightarrow D_{\hat{u}}f=\nabla f\cdot\hat{u}$. Normalize the direction first.
- **B** **steepest ascent** $\Rightarrow$ direction is ∇f, and the rate is $|\nabla f|$. Steepest descent is $-\nabla f$.
- **C** **no change** $\Rightarrow$ any direction perpendicular to ∇f — that is along the level curve.
- **D** **tangent plane to a level surface** $\Rightarrow \nabla F$ is the normal vector.

The gradient answers all four, so compute ∇f first and then read the question again. The single most common lost mark is skipping the normalization in A.

max/min — critical points & the second derivative test

Which hunt is this? — three problems

- **A** **classify critical points** (no region given) $\Rightarrow$ solve $\nabla f=\vec{0}$, then run the D test on each.
- **B** **absolute max/min on a closed region** $\Rightarrow$ collect three sets of candidates: interior critical points, each edge, and every corner. Compare values. No D test needed.
- **C** **optimize with a constraint** $\Rightarrow$ substitute the constraint to remove a variable, or use Lagrange: $\nabla f=\lambda\nabla g$.

THE THREE THINGS YOU SOLVE

1. **The system** $f_x=0$, $f_y=0$ — both zero at the same point. Factor and enumerate cases.
2. **Each boundary edge** — substitute the edge equation into f to get a one-variable Calc 1 problem.
3. **Which quantity is constrained** — read whether the volume or the material is the given one. The roles swap. The recipe does not.

double integrals — rectangular regions

THE PROCESS

1. **Iterate** — write it as two nested single integrals *(Fubini: over a rectangle the order is free)*.
2. **Inner** — freeze the outer variable; treat it like the constant 5.
3. **Outer** — now ordinary Calc 1.
4. **Sanity ladder** — each pass kills one variable. The answer is a pure number.

Step 1 decoded — pick the order

- **A** f **factors** as $g(x)h(y)$ and R is a rectangle $\Rightarrow$ split into a product of two singles. Fastest route.
- **B** f does not factor ($\sin(x+y)$, $x+2y$) $\Rightarrow$ nested passes, ordinary way. Exception: $e^{x+y}=e^xe^y$ does factor.
- **C** both orders legal, one is far easier $\Rightarrow$ scout ten seconds first. For xe^{xy}: dx first needs parts; dy first, $\frac{\partial}{\partial y}e^{xy}=xe^{xy}$ — the integrand is already somebody's derivative.

The only thing you solve here: "does f literally equal $g(x)$ times $h(y)$?" Everything else is Calc 1 twice.

double integrals — general regions

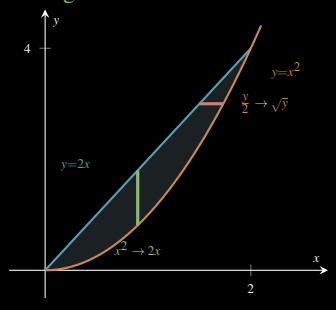
THE PROCESS

1. **Sketch** the region. Limits come from the picture, never from the integrand.
2. **Pick a slice direction** — vertical (Type I) or horizontal (Type II).
3. **Run one test slice**: where does it enter, where does it exit? Those are the inner limits.
4. **How far do slices sweep?** That range is the outer limits — numbers.
5. **Integrate inside out.**

Step 2 decoded — three region shapes

- **A** **Type I**, vertical slices, $dy\,dx$: $a\le x\le b$, $g_1(x)\le y\le g_2(x)$ — bottom curve up to top curve
- **B** **Type II**, horizontal slices, $dx\,dy$: $c\le y\le d$, $h_1(y)\le x\le h_2(y)$ — left curve across to right curve
- **C** **must split** — the entering or exiting curve changes formula partway. Two integrals, added.

Every region is both A and B, but rarely at the same price. Run a test slice both ways and count how many times the boundary formula changes: that count is the number of integrals.



one region, two descriptions: a vertical slice enters at $y=x^2$, exits at $y=2x$ (Type I); a horizontal slice enters at $x=\frac{y}{2}$, exits at $x=\sqrt{y}$ (Type II)

The four things you actually solve

1. WHERE DO THE CURVES MEET? — OUTER LIMITS
set them equal and solve. $x^2=2x \Rightarrow x(x-2)=0 \Rightarrow x=0,2$ · $x^2=2-x^2 \Rightarrow x=\pm 1$

↓

2. WHICH IS TOP / WHICH IS LEFT?
Plug in one interior point. at $x=1$: $2x=2$ vs $x^2=1$ ⇒ the line is on top: $x^2 \le y \le 2x$

↓

3. SOLVE EACH CURVE FOR THE *OTHER* VARIABLE (TYPE II)
$y=x^2 \Rightarrow x=\sqrt{y}$ · $y=2x \Rightarrow x=\frac{y}{2}$ · $y=2-x^2 \Rightarrow x=\pm\sqrt{2-y}$
Both roots survive if the region straddles the axis.

↓

4. DOES THE BOUNDARY CHANGE FORMULA?
run a test slice at two different heights. If they enter/exit on different curves, split there — and the split value is where the two boundary curves cross.

THE PROCESS
1. Set up the region once. The limits serve both integrals.
2. Integrate f over it — the numerator.
3. Integrate 1 over it — the denominator.
4. Divide.

double integrals — polar coordinates

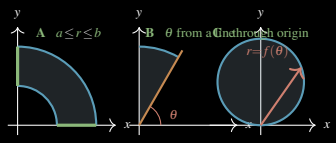
THE PROCESS
1. Region → polar: rays give θ, arcs and curves give r.
2. Integrand → polar: every x and y, no exceptions.
3. Attach the r — $dA=r\,dr\,d\theta$, even when the integrand never asked.
4. Integrate inside out, r first.

Step 1 decoded — three region shapes

A **polar rectangle** — two arcs, two rays ⇒ all four limits constant. Disks, annuli, quarter-disks.

B **wedge cut by a line** — a boundary $y=mx$ is the ray $\theta=\arctan m$. θ comes from the line, not from habit.

C $r=f(\theta)$ — a curve through the origin (cardioid, or a circle not centred there) ⇒ the inner limit carries θ.



The four things you actually solve

1. DOES θ SURVIVE? — CONVERT FIRST, THEN LOOK
if the converted integrand still has $\cos\theta$ or $\sin\theta$, the outer pass is a real trig integral. An x, y or xy always leaves one behind; x^2+y^2 never does.

↓

2. θ FROM A LINE
$\tan\theta=\frac{y}{x}=m \Rightarrow \theta=\arctan m$.

$$y=x \Rightarrow \theta=\tfrac{\pi}{4} \qquad y=\sqrt{3}x \Rightarrow \theta=\tfrac{\pi}{3} \qquad x=0 \Rightarrow \theta=\tfrac{\pi}{2}$$

↓

3. OFF-CENTRE CIRCLE → $r=f(\theta)$
substitute $x^2+y^2=r^2$ and divide by r:

$$x^2+y^2=2ay \Rightarrow r^2=2ar\sin\theta \Rightarrow r=2a\sin\theta \ (\theta: 0 \to \pi)$$

$$x^2+y^2=2ax \Rightarrow r=2a\cos\theta \ (\theta: -\tfrac{\pi}{2} \to \tfrac{\pi}{2})$$

(complete the square to confirm the centre: $x^2+(y-a)^2=a^2$)

↓

4. CLOSE THE SHADOW — SET ROOF = FLOOR
for a volume, the solid's edge is where the two surfaces meet:

$$2y=x^2+y^2 \Rightarrow r=2\sin\theta \qquad x^2+y^2=4 \Rightarrow r=2$$

triple integrals — rectangular coordinates

THE PROCESS — EVERY TRIPLE INTEGRAL, NO EXCEPTIONS
1. Inner variable — pick the one trapped between two surfaces.
2. Solve both of those surfaces for that variable ⇒ inner limits.
3. Project the solid onto the plane of the other two ⇒ the **shadow**.
4. Read the shadow as a §5.2 Type I/II region ⇒ middle & outer limits.
5. Integrate inside out — each pass kills one variable.

Step 1 decoded — which variable goes inside?

Read the list of bounding surfaces and match one of three shapes:

A two surfaces are $z=f(x,y)$ — a roof and a floor ⇒ **inner** $=z$, project onto xy

B a surface never mentions one variable ($x=z^2$ has no y) ⇒ that variable is free: it is **inner**, project onto the plane of the other two

C a vertical wall $x^2+y^2=a^2$ ⇒ **inner** $=z$, shadow is a disk — go **cylindrical** (next section)

(A is the default. B is the one people miss — it is why "project onto xy" is a habit, not a rule.)

The four things you actually solve

Every limit in every problem comes from one of these. Nothing else.

1. WHICH SURFACE IS ON TOP?
Plug in one interior point and compare. No algebra — and skipping it flips your sign.
for example, $z=x^2$ vs $z=x+2$ at $x=0$: $0<2$ ⇒ $x^2 \le z \le x+2$

↓

2. SOLVE A SURFACE FOR THE INNER VARIABLE
Rearrange one equation. A curved face needs a real inversion.

$$x+y+z=1 \Rightarrow z=1-x-y \qquad x=z^2 \Rightarrow z=\sqrt{x}$$

$$z=1-x^2 \Rightarrow x=\sqrt{1-z} \qquad x^2+y^2+z^2=2z \Rightarrow \rho=2\cos\varphi$$

↓

3. CLOSE THE SHADOW — SET TWO BOUNDARIES EQUAL
When the stated fences don't close the region, the surfaces themselves close it. Set them equal, solve, and those roots are outer limits.

$$x^2=2-x \Rightarrow x=1,\,-2 \qquad z^2=3-2z \Rightarrow z=1 \ \textit{(reject } z=-3\textit{)}$$

↓

4. NON-EMPTY CHECK — IT *IS* THE NEXT LIMIT
If the inner run is $g \le x \le h$, demanding $h \ge g$ produces the middle limit. Easy to skip. Skipping it loses a whole limit.

$$\sqrt{1-z} \ge y \Rightarrow 1-z \ge y^2 \Rightarrow z \le 1-y^2$$

Seeing the solid — four moves before any integral

Most lost time here is spent trying to draw one good 3D picture. Do not. Draw flat pictures instead, in this order:

1. SPOT THE EXTRUDED SURFACES — THE BIGGEST SIMPLIFICATION
A surface missing a variable is a **cylinder**: it extrudes along that axis, unchanged.
$z=1-x^2$ has no y ⇒ a parabolic sheet running the length of the y-axis. $y=1-x$ has no z ⇒ a slanted wall running straight up.
Two extruded surfaces missing *different* variables is what makes a solid hard.

↓

2. DRAW THE THREE TRACES
Set $x=0$, then $y=0$, then $z=0$. Each gives a flat curve or region in a coordinate plane. Three flat pictures beat one bad 3D sketch.

↓

3. FIND THE CORNERS
Solve the surfaces in pairs. Corners pin the solid down and catch a mis-drawn face immediately.

↓

4. GET ALL THREE SHADOWS AT ONCE
Project onto xy, xz and yz. Every one of the six orderings reads its outer and middle limits off one of these three pictures. Compute them once and the reordering drill becomes bookkeeping.

THE LAST TWO ORDERINGS SPLIT — AND HERE IS WHY
put x innermost. It is capped by both walls at once:

$$x \le 1-y \quad \text{and} \quad x \le \sqrt{1-z} \qquad \Rightarrow \qquad x \le \min\left(1-y,\ \sqrt{1-z}\right)$$

A *min* in a limit always means a split. Find where the two caps trade places:

$$\sqrt{1-z}=1-y \Rightarrow 1-z=(1-y)^2 \Rightarrow z=2y-y^2$$

That curve cuts the yz square in two.

coordinate conversions — the full table

THE ONLY THREE FACTS TO MEMORIZE
1. $r=\rho\sin\varphi$ — the shadow of ρ on the floor.
2. $z=\rho\cos\varphi$ — the height of ρ.
3. θ **never changes** — it is the same angle in cylindrical and spherical.
Everything below follows from these three plus $x=r\cos\theta$, $y=r\sin\theta$.

All six directions

direction	formulas
rect → cyl	$r=\sqrt{x^2+y^2}$, $\theta=\arctan\left(\frac{y}{x}\right)$, $z=z$
cyl → rect	$x=r\cos\theta$, $y=r\sin\theta$, $z=z$
rect → sph	$\rho=\sqrt{x^2+y^2+z^2}$, $\theta=\arctan\left(\frac{y}{x}\right)$, $\varphi=\arccos\left(\frac{z}{\rho}\right)$
sph → rect	$x=\rho\sin\varphi\cos\theta$, $y=\rho\sin\varphi\sin\theta$, $z=\rho\cos\varphi$
cyl → sph	$\rho=\sqrt{r^2+z^2}$, $\theta=\theta$, $\varphi=\arccos\left(\frac{z}{\sqrt{r^2+z^2}}\right)$
sph → cyl	$r=\rho\sin\varphi$, $\theta=\theta$, $z=\rho\cos\varphi$
rect → polar	$r=\sqrt{x^2+y^2}$, $\theta=\arctan\left(\frac{y}{x}\right)$ *(2D — same quadrant fix)*
polar → rect	$x=r\cos\theta$, $y=r\sin\theta$

The 2D pair is cylindrical with the z dropped — polar is cylindrical, flattened. A worked quadrant fix lives in the Trigonometry section; the same arctan trap below applies. The cyl/sph pair is the easiest — cylindrical and spherical differ only in how they name the floor distance and the height. **Volume elements**: $dV=dz\,dy\,dx$, $dV=r\,dz\,dr\,d\theta$, $dV=\rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta$.

TWO TRAPS IN THE ANGLES

arctan returns only QI and QIV. If $x<0$, add π to θ. Sketch the point and check the quadrant every time.

Use arccos for φ, not arctan$\left(\frac{r}{z}\right)$. With $r=3$, $z=-4$: $\arccos\left(\frac{-4}{5}\right)=143°$ ✓, but $\arctan\left(\frac{3}{-4}\right)=-37°$ fails — it lands below the xy-plane, where φ can never go.

Converting surfaces — *the row you will use most*

Points are rare on an exam. Surfaces are the real job: you convert a boundary to read a limit off it.

rectangular	cylindrical	spherical	use
x^2+y^2	r^2	$\rho^2\sin^2\varphi$	
$x^2+y^2+z^2$	r^2+z^2	ρ^2	
z	z	$\rho\cos\varphi$	
cylinder $x^2+y^2=a^2$	$r=a$	$\rho\sin\varphi=a$	cyl
plane $z=c$	$z=c$	$\rho\cos\varphi=c$	either
cone $z=a\sqrt{x^2+y^2}$	$z=ar$	$\varphi=\arctan\frac{1}{a}$	sph
sphere $x^2+y^2+z^2=a^2$	$r^2+z^2=a^2$	$\rho=a$	sph
sphere $x^2+y^2+z^2=2az$	$r^2+z^2=2az$	$\rho=2a\cos\varphi$	sph
paraboloid $z=a(x^2+y^2)$	$z=ar^2$	$\rho=\frac{\cos\varphi}{a\sin^2\varphi}$	cyl

The use column names the system where that surface is simplest — integrate there. A cone is one *number* in spherical (φ is constant). A paraboloid stays ugly in spherical and is clean in cylindrical.

The 45° cone: $a=1$ gives $\varphi=\arctan 1=\frac{\pi}{4}$. Steeper cone $z=2r$: $\varphi=\arctan\frac{1}{2}\approx 26.6°$ — not a nice angle, which is why textbook cones are almost always $z=\sqrt{x^2+y^2}$.

triple integrals — cylindrical coordinates

THE PROCESS — IT IS THE TRIPLE PROCESS, SHADOW IN POLAR

1. Inner $=z$ — roof and floor, solved for z *(variant A or C of the triple table)*.
2. Project onto xy ⇒ the shadow, then read it in polar (§5.3's three shapes).
3. Attach the r — $dV=r\,dz\,dr\,d\theta$.
4. Integrate inside out: z, then r, then θ.

triple integrals — spherical coordinates

THE PROCESS

1. Convert every surface to ρ,φ,θ — use $x^2+y^2+z^2=\rho^2$ and $z=\rho\cos\varphi$.
2. ρ **innermost**, φ middle, θ outer — almost always.
3. Attach $dV=\rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta$.
4. Integrate inside out. If all six limits are constants, the integral separates into three.

Step 1 decoded — three region shapes

A **ball / spherical shell** ⇒ ρ between constants, $\varphi:0\to\pi$, $\theta:0\to 2\pi$. Fully separable.

B **cone** $z=a\sqrt{x^2+y^2}$ ⇒ it is just a φ limit: $\tan\varphi=\frac{1}{a}$. The 45° cone $z=\sqrt{x^2+y^2}$ is $\varphi=\frac{\pi}{4}$.

C **sphere through the origin** $x^2+y^2+z^2=2az$ ⇒ $\rho=2a\cos\varphi$ — the ρ-limit carries φ, and the integral no longer separates.

Which coordinate system? — the decision guide

- **Cartesian**: boxes, planes, tetrahedra — straight-edged regions.
- **Polar / cylindrical**: circles in 2D; axis symmetry in 3D — cylinders, cones, paraboloids. The tell is x^2+y^2.
- **Spherical**: point symmetry — spheres, balls, cones through the origin. The tell is $x^2+y^2+z^2$.
- **Your own** u,v: none of the above fits, but the integrand names a combination — $e^{(x-y)/(x+y)}$, $(x+2y)^{10}$, a slanted parallelogram. Build the map, take the Jacobian.
- The question also gets a vote: "find the mass" means the integrand is a density, not 1 — same region, same coordinates, different f.
- The integrand gets a vote: $e^{x^2+y^2}$ forces polar even over an ugly region — a rescue-worthy integrand outranks a pretty region.

mass, center of mass & moments

THE PROCESS — NOTHING NEW, ONE NEW INTEGRAND

1. Set up the region exactly as before — same shadow, same limits, same coordinates.
2. Swap the integrand: $f=1$ gives volume, $f=\delta$ gives mass, $f=x\delta$ gives a moment.
3. Use symmetry first — if δ never mentions a variable and the region is symmetric in it, that coordinate of the balance point is the geometric middle, for free.
4. Divide by the mass — never by the volume.

change of variables — where every Jacobian comes from

THE PROCESS

1. Let the integrand choose u **and** v — name the combinations that appear in it.
2. Invert: solve for x and y in terms of u,v. The Jacobian needs them this way round.
3. Map every boundary edge into the uv-plane ⇒ the new region S.
4. Take the determinant, then its absolute value.
5. Integrate over S with the extra factor.

Discrete I — Logic, Sets & Proofs

algebra refreshers

Which method? — decide by counting terms

1. **GCF first, always.** Pull out the biggest factor every term shares. Check: multiply back to confirm.
2. **Count the terms:**
   - 2 terms → diff. of squares a^2-b^2, or sum/diff. of cubes.
   - 3 terms → trinomial. Perfect square? Else do the p,q search below.
   - 4 terms → grouping.

proof strategies

THE PROCESS
1. Read the shape of the claim — $p \to q$, "for all", "there exists", or a set equality.
2. Try to break it first. One counterexample ends the job. Do this before proving anything.
3. Pick the technique from the table below.
4. Write the first and last lines before the middle — what you assume, and what you must reach.
5. Close the gap between them.

Step 3 decoded — which technique?

technique	reach for it when
Direct	p unpacks cleanly into q
Contrapositive	$\neg q$ is easier to handle than p — often when q says "not", or denies existence
Contradiction	irrationality, infinitude, uniqueness
Cases	a definition splits: absolute value, parity, sign
Induction	the claim is indexed by $\mathbb{N}$ and the pattern propagates
Strong induction	recurrences, or splitting into smaller pieces
Counterexample	you suspect the claim is false

THE TWO LINES THAT DO THE WORK
Opening — unpack the hypothesis into algebra: "n is even, so $n=2k$ for some integer k."
Closing — state the goal in the same algebra: "so $n^2=2m$ for some integer m, and n^2 is even."
Write both before the middle. Most stuck proofs are stuck because the goal was never written down.

Discrete II — Counting, Recurrences & Graphs

combinatorics

THE PROCESS — TWO QUESTIONS DECIDE EVERYTHING
1. Does order matter? Swap two chosen items — does the result change?
2. Is repetition allowed? May an item be picked twice?
3. Read the count off the table.
4. Check for overlap or symmetry — then correct with inclusion-exclusion or the division rule.

Steps 1–2 decoded — the counting table

	no repetition	repetition allowed
order matters	$P(n,k)=\dfrac{n!}{(n-k)!}$	n^k
order does not	$C(n,k)=\binom{n}{k}$	$\binom{n+k-1}{k}$

The bottom-right cell is **stars and bars**. Link: $P(n,k)=C(n,k)\cdot k!$ — a permutation is a combination times its orderings.

Step 4 decoded — three corrections

A **"at least one"** ⇒ count the complement and subtract from the total. Counting it directly means counting every case.
B **overlapping sets** ⇒ inclusion-exclusion: $|A\cup B|=|A|+|B|-|A\cap B|$
C **identical items or a symmetry** ⇒ division rule: count as if distinct, then divide by the duplicates.